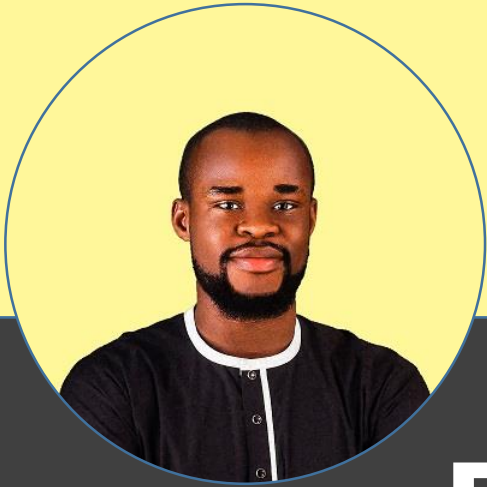




Business Intelligence Analytics

A large scale business is celebrating their anniversary. They needed information about how the business is doing so as to adequately compensate loyal customers. They also discussed concerns of increasing return rates and wanted to know why.



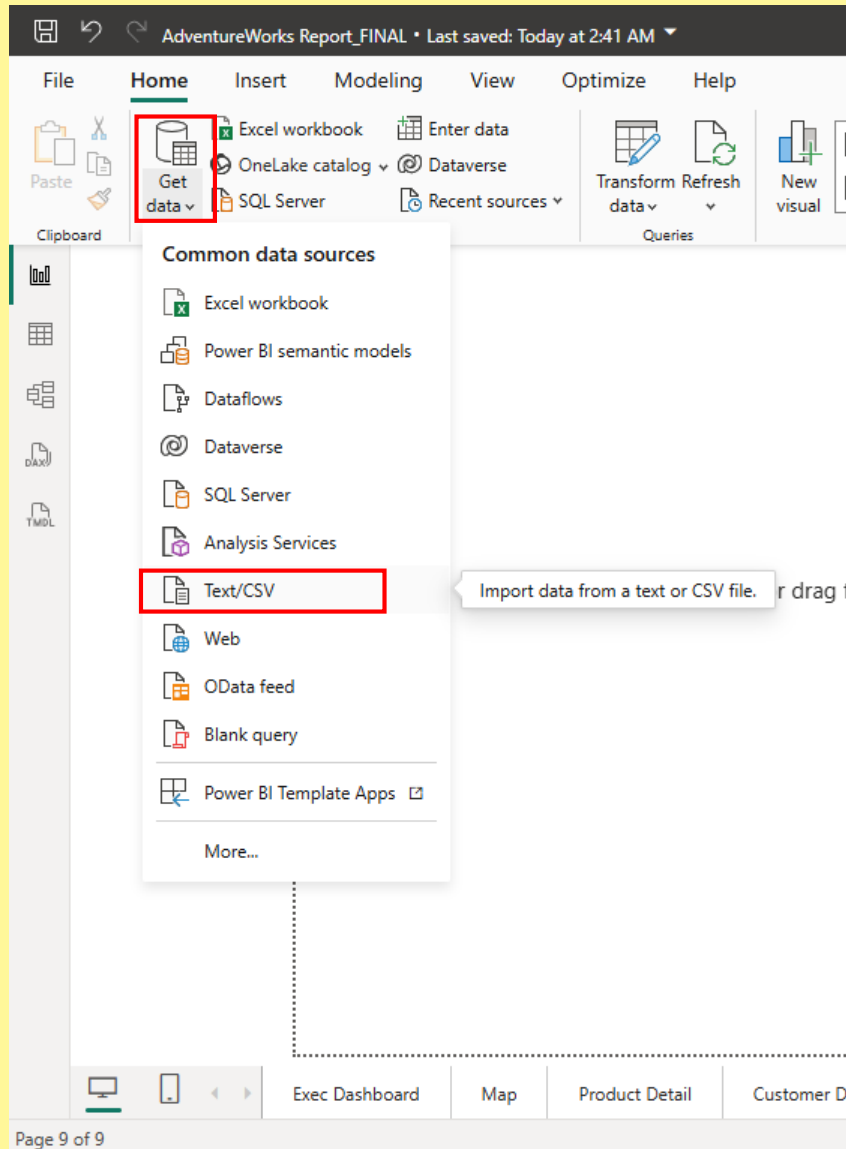
Overview

This project involved developing an interactive Power BI dashboard to analyze sales performance and key business KPIs. By consolidating revenue, profit, orders, customer, product, and regional data into a single reporting solution, the dashboard enables stakeholders to monitor performance, identify trends, and make informed, data-driven decisions.

Business Problems/Questions(EDA)

1. What is the business performance based on the KPI (Revenue, Profit, Orders, return Rate)?
2. Which Customer has invested the most over the years?
3. Which region has produced the most revenue?
4. What products are the most sold and the most returned?
5. What category of products has the most orders?
6. Predict revenue with a 10% increase in price.
7. How are each product performing?
8. What are the top 10 products by orders?
9. What is the revenue trend over the years?
10. What is the revenue, profit and returns performances compared with previous months
11. What is the average revenue generated per customer?
12. What is the total order by the top 3 Income level and occupation?
13. Why is there an increase in return rate?

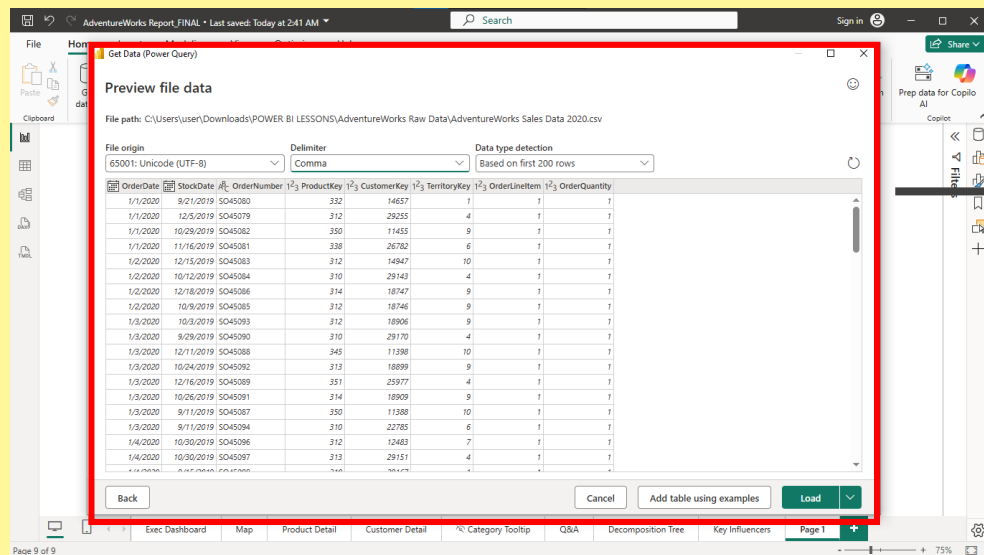
1. Importing the Files



There are a number of ways to get your data into the work space including from databases, but for the sake of the demonstration we will be importing Text/CSV files already extracted from the data base using SQL.

Clicking on the get data icon brings a pop up that prompts you to choose import style and then locate the file on your computer

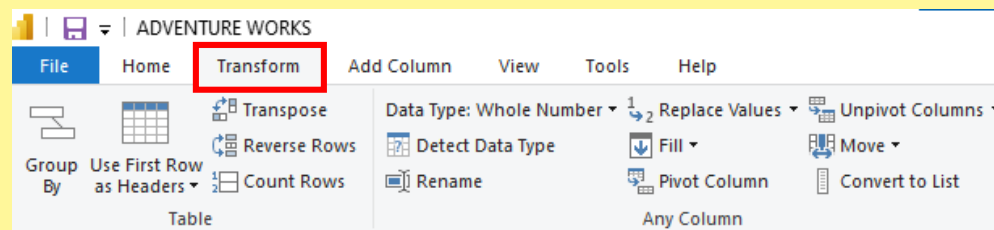
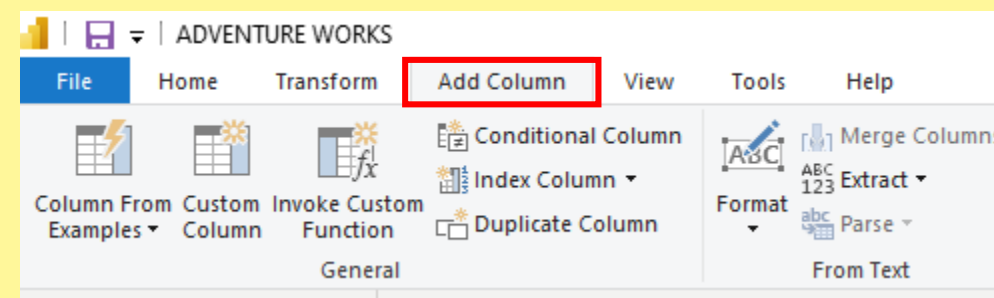
2. Data Cleaning (Power Query)



After selecting your data cleaning is important is this happens through the Power Query, a powerful tool that allows irregular, missing values to be tackled and also allows the creation of calculated columns.

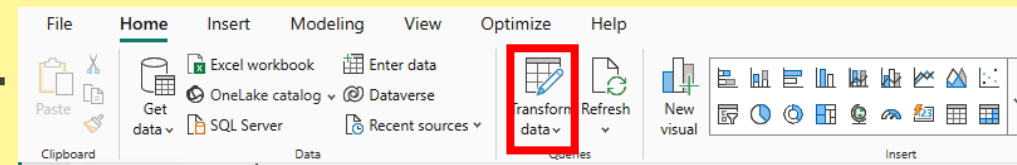
The add column and transform tabs and parts of the power query tool that allows you modify your data before creating relationships. Following the need of the business there is usually a clear idea on tables that need to be included

N/B: these are great tools but it is important not to overuse them cause they can make the file really heavy creating a lot new and redundant data. DAX measures are a preferred way. So calculated columns should be used only when necessary



3. Previewing Data Tables after cleaning

After cleaning the data its important to carefully observe and walk through in case anything was missed. The Transform data tab allows you access the power query



This tab shoesh all the tables loaded into the power query

A screenshot of the Microsoft Power Query 'Transform' tab interface. On the left, a list of queries is shown, with 'Customer Lookup' selected and highlighted by a red box. In the center, a data preview table is displayed with columns: CustomerKey, Prefix, FirstName, LastName, BirthDate, and MaritalStatus. The table contains 26 rows of data. On the right, the 'APPLIED STEPS' pane is visible, listing various transformations applied to the data, including 'Source', 'Promoted Headers', 'Changed Type', 'Removed Errors', 'Filtered Rows', 'Capitalized Each Word', 'Full Name', 'Duplicated Email Column', 'Extracted Domain', 'Renamed Columns1', 'Extracted Text Before Delimiter', 'Replaced Value', and 'Capitalized Each Word1'. The 'Capitalized Each Word1' step is currently selected and highlighted by a red box. The top ribbon shows tabs for File, Home, Transform, Add Column, View, Tools, and Help. The bottom status bar indicates '15 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

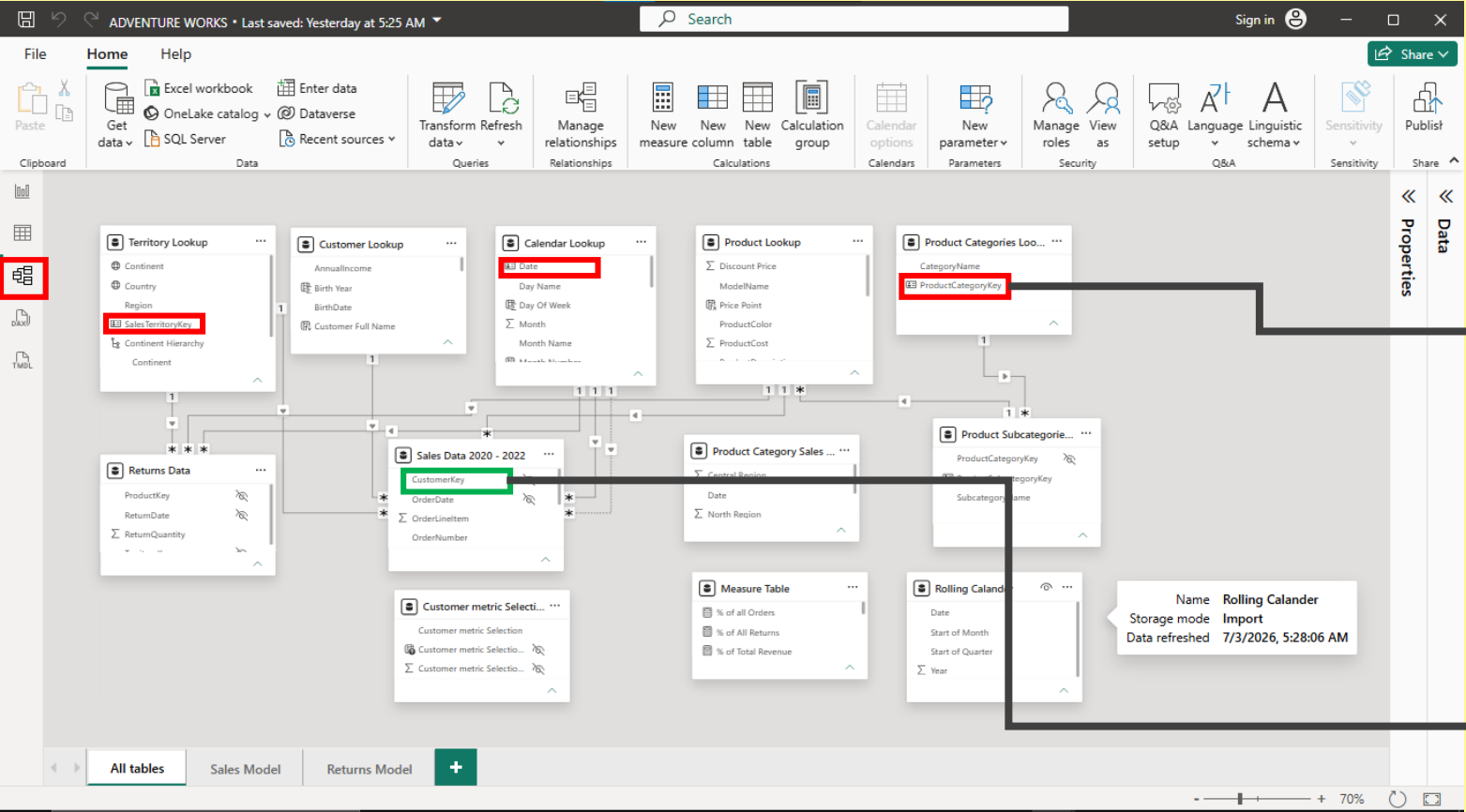
The applied steps tab shows all steps taken in transforming the data within the power query

This tab displays the current selected table

4. Data Modeling

This is a vital part of the analysis it involves creating relationships between previously independent tables allowing them to communicate without merging or combining them. This relationships are created through **Primary and Foreign keys**.

This tab opens the data modelling view



Columns in red are Primary keys which are unique identifiers of a table

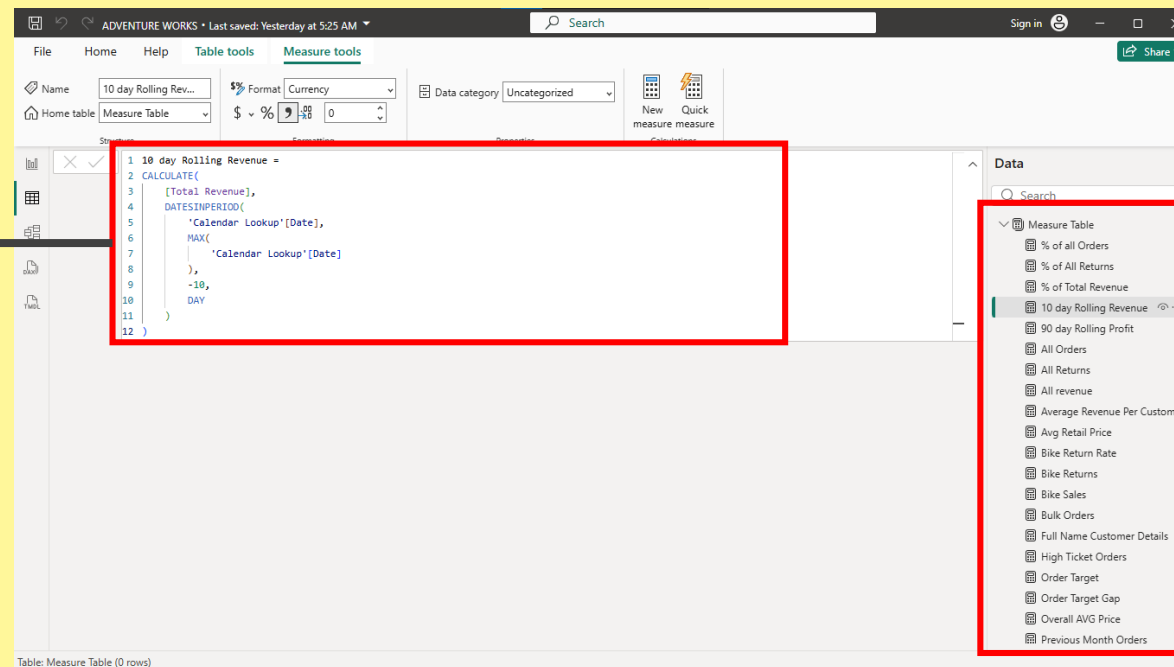
This is a foreign key it is a key that doesn't have unique entities

N/B: The pattern for connect a primary key to a foreign key is “one to many”

5. DAX

DAX measures are one of the amazing tools that allows a lot of computations and calculations without the need to create redundant calculated columns. They are a big part of the analysis process.

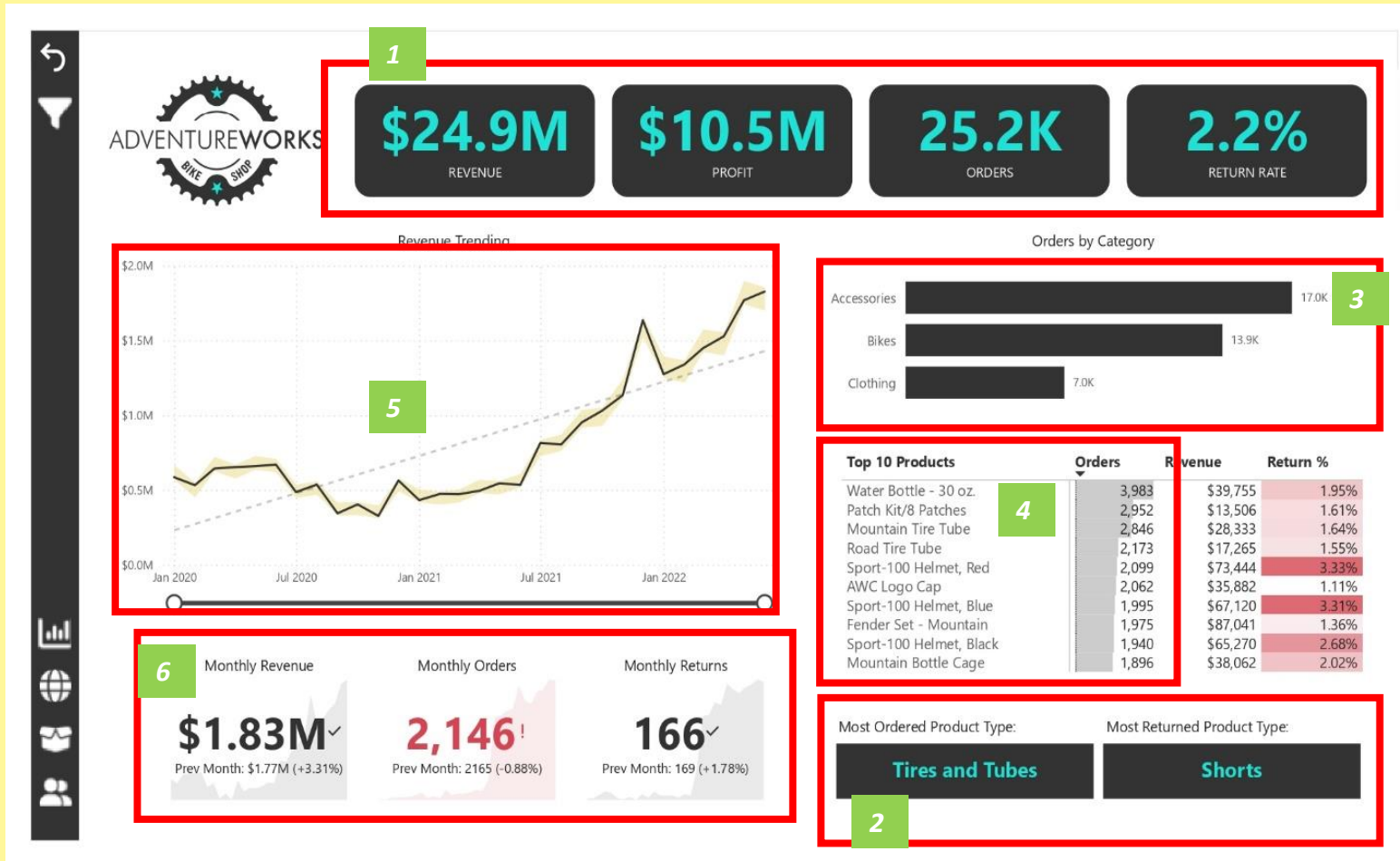
This showcases a sample of a dax measure created to calculate a 10 day Rolling revenue



This shows a table created to house all the DAX measures created in the course of this analysis

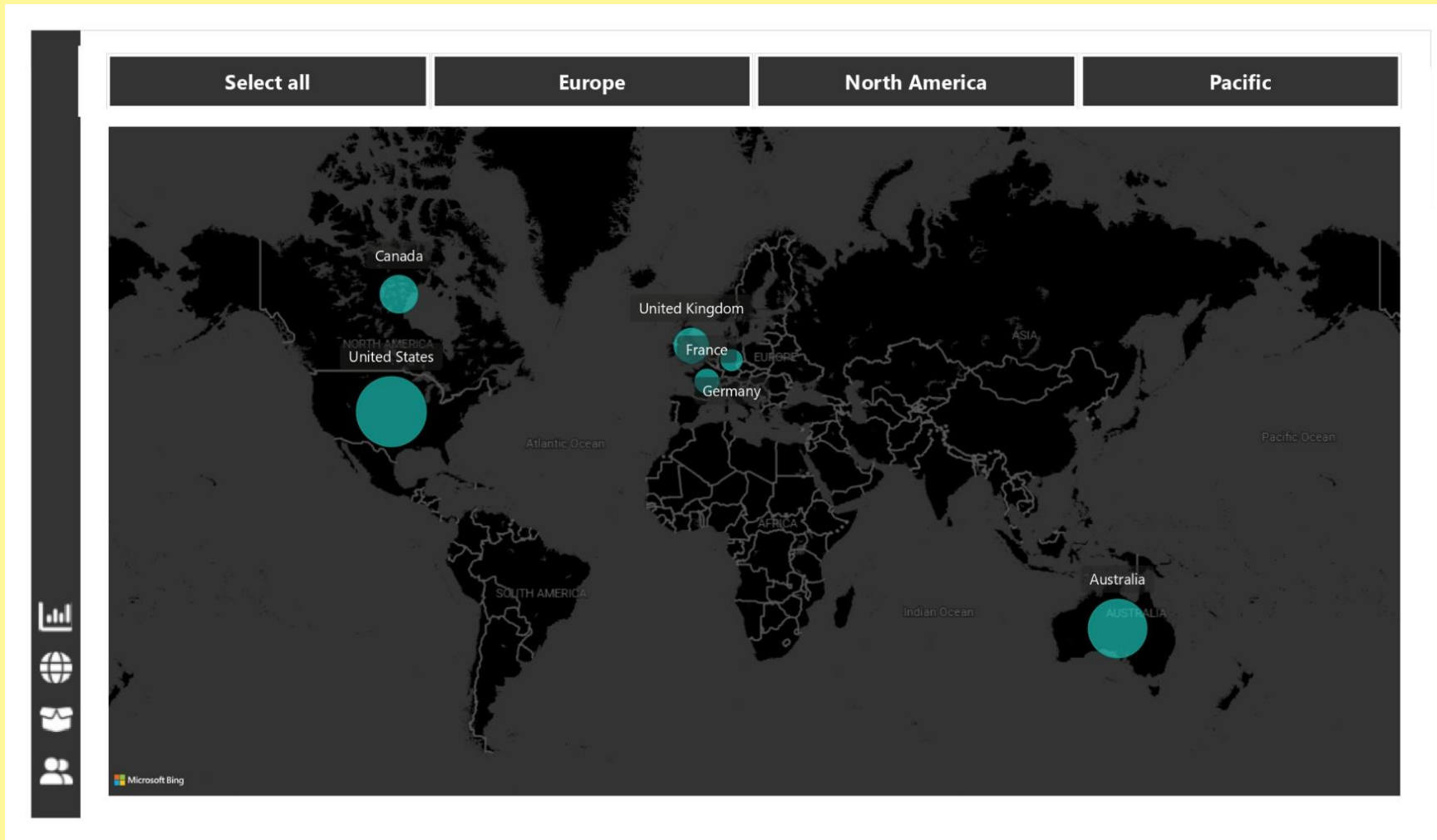
**Now we answer the business
questions using Data Visualization
(EDA)**

6. Data Visualization



1. What is the business performance based on the KPI (Revenue, Profit, Orders, Return Rate)?
2. What products are the most sold and the most returned? And why?
3. What category of products has the most orders?
4. What are the top 10 products by orders?
5. What is the revenue trend over the years?
6. What is the revenue, profit and returns performances compared with previous months

Data Visualization

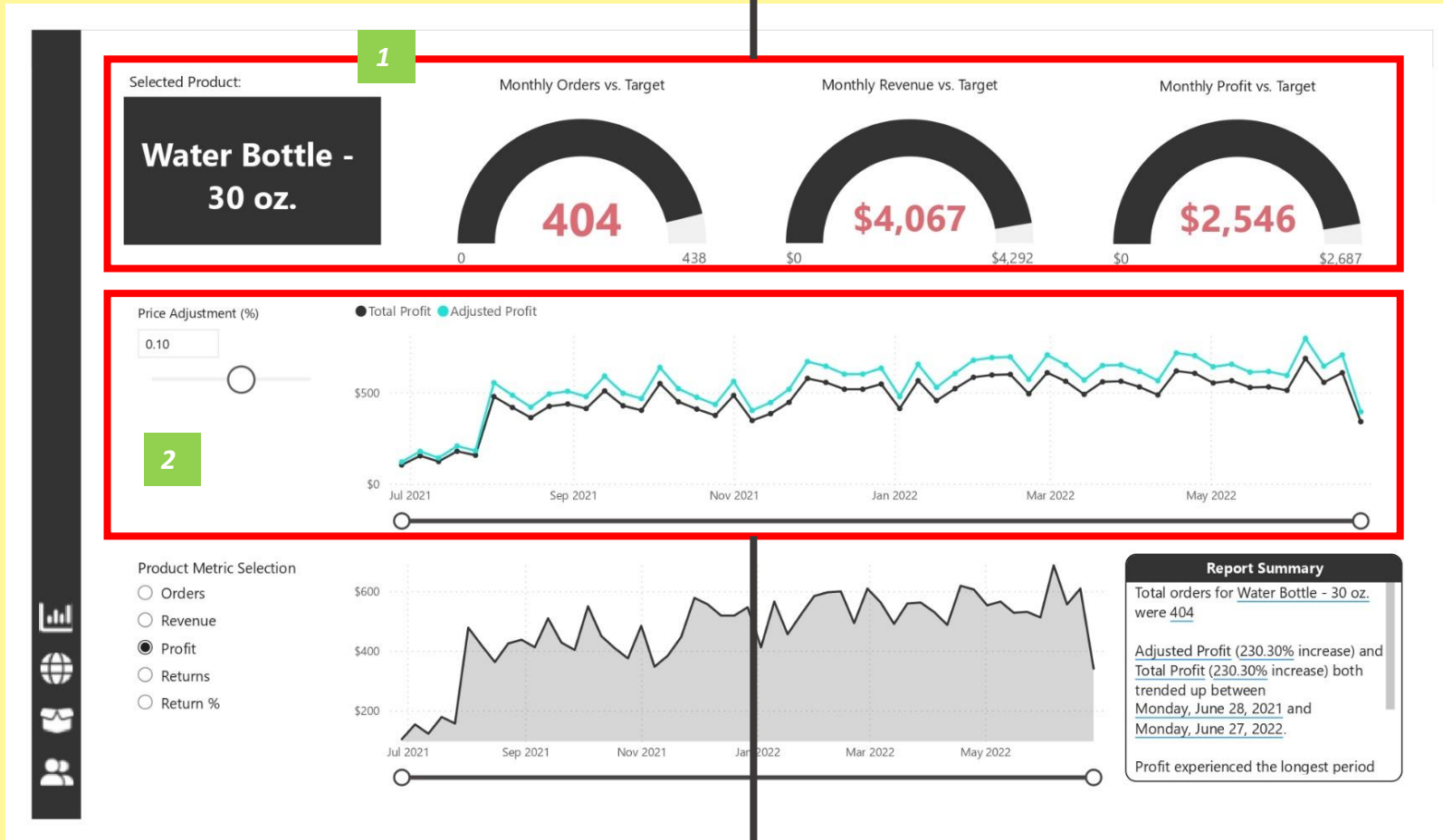


1. Which region has produced the most revenue?

The bubble size represents order quantity

Data Visualization

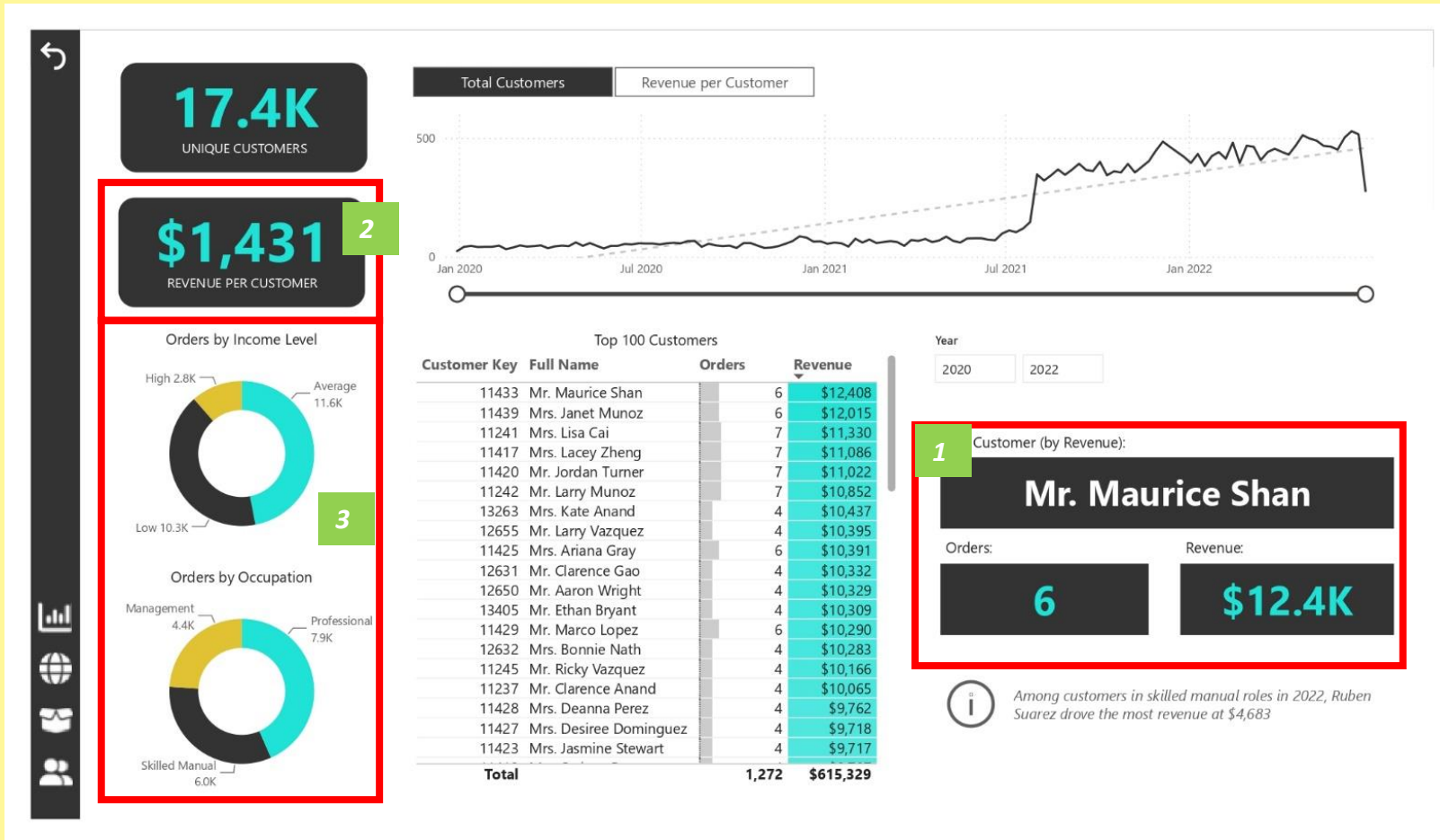
This shows the comparison of orders, profits and revenue between currently monthly amount and target amounts of each product.



1. Predict revenue with a 10% increase in price.
2. How are each product performing?

Slider on the far left allows us to view a predicted profit mount if prices were adjusted by a certain percent

Data Visualization



1. Which Customer has invested the most over the years?
2. What is the average revenue generated per customer?
3. What is the total order by the top 3 Income level and occupation?

Data Visualization

Start of Month	Orders	Return Rate
1/1/2020	184	2.17%
2/1/2020	165	2.42%
3/1/2020	198	4.55%
4/1/2020	204	6.86%
5/1/2020	206	5.34%
6/1/2020	212	1.89%
7/1/2020	247	1.21%
8/1/2020	278	2.16%
9/1/2020	196	1.02%
10/1/2020	223	4.93%
11/1/2020	191	2.62%
12/1/2020	326	3.99%
1/1/2021	242	3.31%
2/1/2021	267	3.00%
3/1/2021	266	3.01%
4/1/2021	290	1.72%
5/1/2021	329	3.04%
6/1/2021	312	2.56%
7/1/2021	506	2.30%
8/1/2021	1543	2.01%
9/1/2021	1568	2.04%
10/1/2021	1639	2.14%
11/1/2021	1677	2.15%
12/1/2021	2056	2.06%
1/1/2022	1811	2.25%
2/1/2022	1758	2.26%
3/1/2022	1962	2.18%
4/1/2022	1997	2.14%
5/1/2022	2165	2.06%
6/1/2022	2146	2.02%

1. Why is there an increase in return rates?

Return rates are relatively same in recent times but orders have skyrocketed

Conclusion

- I've heard people talk about the power of DAX measures, I didn't quite understand why,
- but doing this project, I made use of it 90% of the queries. • The range of flexibility it allowed, making possible the exploration of data in ways that calculated columns would struggle at.
- The business didn't have an increase in return rates as the rates remained relatively constant in recent times but there was a huge increase in orders